model wasn't suitable for Ubuntu—it cannot start properly. Moreover, newer versions of Ubuntu come without any POWER/PowerPC architecture support at all, unfortunately.

I checked if the most recent release (6.0.3) of Ubuntu's parent distribution, Debian, could boot. It wasn't able to start—it drops directly into OpenFirmware. Another failure! Well, let's not despair. Quite possibly, the Debian community will fix it.

Oddly enough, a completely amateurish project called Crux PPC [7] has a fully working distribution—Crux PPC 2.7a, which not only loads on Power hardware, but also provides video output via correctly-set framebuffer mode. None of the other distributions mentioned can boast of that! All provided only a serial-line text mode. In general, Crux PPC runs as a LiveCD and has no installer, so I advise professionals to use it. Again, this topic is out of the scope of this article, so I will leave it for later.

Performance

As I mentioned a bit earlier, IBM products are one level, or a 'head above' their competitors. Comparing the very different Power and x86 architectures is hard, because there are no objective criteria for such a comparison. However, from an ordinary user's perspective, we can try the 7Zip archiver in its benchmark mode. This program is completely open source, so anyone can compile it.

Let us compare IntelliStation based on the POWER5+ processor with 2 cores and 4 GB RAM, manufactured in 2005, with an Intel Core 2 Duo (2 cores with hyper-threading) and 4 GB RAM, but released three years later (2008). First, the IntelliStation:

```
linux:~/src/p7zip_9.20.1/bin # ./7za b

7-Zip (A) 9.20  Copyright (c) 1999-2010 Igor Pavlov  2010-11-18
p7zip Version 9.20 (locale=en_US.UTF-8,Utf16=on,HugeFiles=on,2
CPUs)

RAM size:    3632 MB,  # CPU hardware threads:   2
RAM usage:    425 MB,  # Benchmark threads:      2

Dict        Compressing          |        Decompressing
      Speed Usage    R/U Rating  |  Speed Usage    R/U Rating
       KB/s     %   MIPS   MIPS  |   KB/s     %   MIPS   MIPS

22:    1931    99   1891   1878  |  24309   100   2198   2194
23:    1871   100   1916   1907  |  24132   100   2212   2209
24:    1816    99   1964   1953  |  23917   100   2221   2219
25:    1780    99   2043   2032  |  23525   100   2213   2212
----------------------------------------------------------------
Avr:           99   1954   1942               100   2211   2209
Tot:          100   2082   2076
```

Next, let us test the Core 2 Duo system:

```
Intel(R) Core(TM)2 Duo CPU     E6550  @ 2.33GHz
7-Zip 9.04 beta  Copyright (c) 1999-2009 Igor Pavlov  2009-05-30
p7zip Version 9.04 (locale=en_US.UTF-8,Utf16=on,HugeFiles=on,2
CPUs)

RAM size:    4006 MB,  # CPU hardware threads:   2
RAM usage:    425 MB,  # Benchmark threads:      2

Dict        Compressing          |        Decompressing
      Speed Usage    R/U Rating  |  Speed Usage    R/U Rating
       KB/s     %   MIPS   MIPS  |   KB/s     %   MIPS   MIPS

22:    2588   138   1831   2518  |  33730   165   1850   3045
23:    2532   136   1903   2580  |  32265   159   1853   2954
24:    2588   142   1958   2783  |  32828   165   1843   3046
25:    2540   144   2011   2900  |  30299   155   1838   2849
----------------------------------------------------------------
Avr:          140   1925   2695               161   1846   2974
Tot:          150   1886   2834
```

As you can see, the performance doesn't differ much. Even a slightly outdated IBM station can compete with a relatively new x86 product.

Despite the fact that equipment based on the Power processor is marketed by IBM as a corporate powerhorse for commercial operating systems like AIX and IBM i (produced by IBM itself), we have seen that it can also be used with Linux. Thus, virtually the entire software stack that successfully works on x86 can also be run on Power— Apache, Java, WebSphere, MySQL, etc. Taking into account the fact that IBM hardware usually comes with unique technologies like the hardware management console and the hardware virtualisation hypervisor, you can easily create an effective solution by combining the best features from both the hardware and software sides—the corporate and the public world. END

Resources

[1] http://en.wikipedia.org/wiki/HP_Integrated_Lights-Out
[2] http://en.wikipedia.org/wiki/IBM_Hardware_Management_Console
[3] http://en.wikipedia.org/wiki/LPAR
[4] http://en.wikipedia.org/wiki/DLPAR
[5] http://www.datasheetarchive.com/indexdl/Datasheets-SW3/DSASW0048000.pdf
[6] http://www.webopedia.com/TERM/F/Flexible_Service_Processor.html
[7] http://cruxppc.org

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